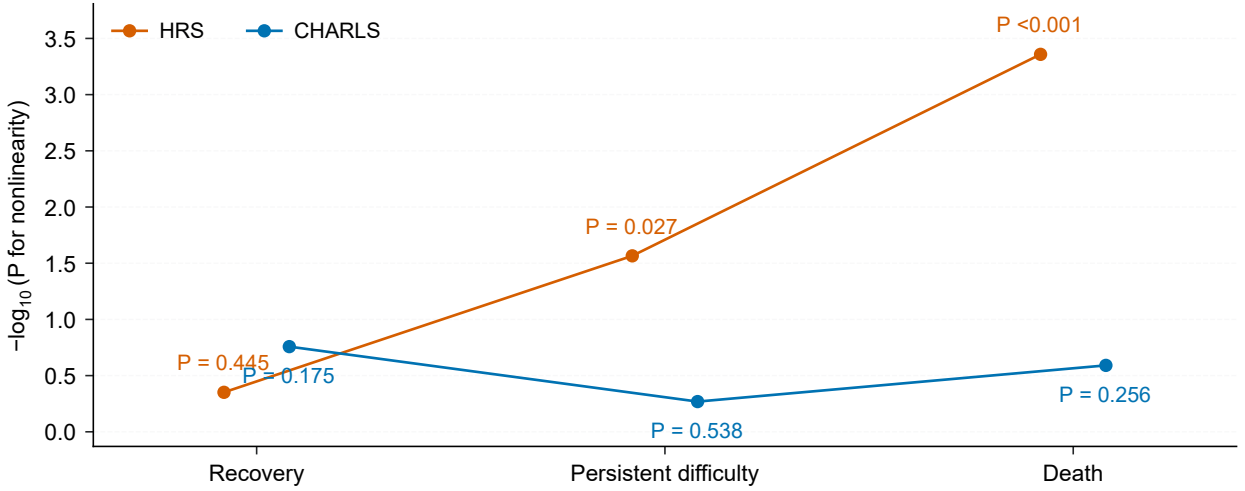A Nonlinearity tests for all observed next-wave states